

SMART FACTORY IoT MONITORING SYSTEM

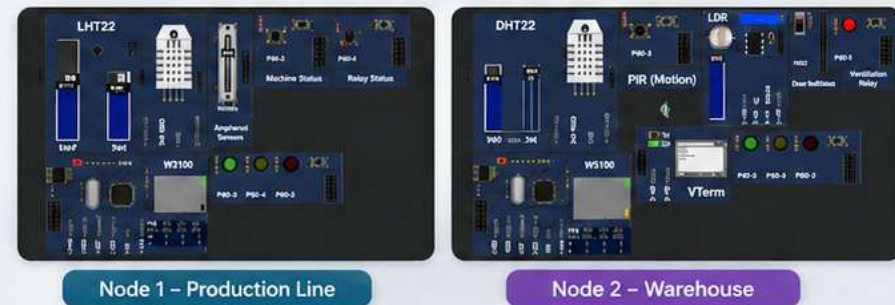
Real-Time Industrial Monitoring & Intelligent Automation
using ThingsBoard Cloud



Smart Factory Overview Dashboard (ThingsBoard Cloud)



PicsimLab Simulation (Dual Node Setup)



PLATFORM
PicsimLab +
ThingsBoard Cloud



PROTOCOL
MQTT



ARCHITECTURE
Dual Node
Industrial IoT



KEY HIGHLIGHTS
Real-Time Monitoring,
Automation, Alerts & Control



PRESENTED BY
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OBJECTIVE

Why We Built This System?

To build a smart, reliable and automated monitoring system for industrial environments using IoT and cloud technologies.



THE PROBLEM



No Real-Time Monitoring

Critical parameters are not monitored continuously.



Delayed Fault Detection

Issues are detected late, leading to equipment damage.



Manual Supervision

Continuous human monitoring is labour intensive and error-prone.



Inefficient Operations

Downtime increases and overall productivity decreases.



Lack of Centralized Data

Data is scattered, no historical insights for analysis.



OUR OBJECTIVE



Develop a cost-effective IoT based monitoring system that enables **real-time data acquisition**, **automation** and **remote monitoring** of industrial environments.

THE SOLUTION



Real-Time Data Monitoring

Live data from multiple sensors visualized on dashboards.



Automated Control

Devices are controlled automatically based on threshold rules.



Remote Access & Control

Monitor and control remotely via cloud (RPC).



Instant Alerts & Alarms

Immediate notifications for critical events.



Centralized Data & Analytics

All data stored in cloud for trend analysis and insights.



ICON MEANING



Real-Time Monitoring



Fault / Issue



Manual Process



Inefficiency / Downtime



Data / Records



Objective



Solution / Benefit



Alert / Alarm



Cloud System

HARDWARE & TECHNOLOGIES

Components that Power the Smart Factory IoT Monitoring System



HARDWARE COMPONENTS



Arduino Uno

Microcontroller processing unit



DHT22

Temperature & Humidity Sensor



LM35

Analog Temperature Sensor



PIR Sensor (HC-SR501)

Motion Detection Sensor



LDR

Light Intensity Sensor



Reed Switch

Door Open/Close Detection



Relay Module

Actuator for Load Control



Ethernet Module (W5100)

Network Interface (Internet)



LED Indicators

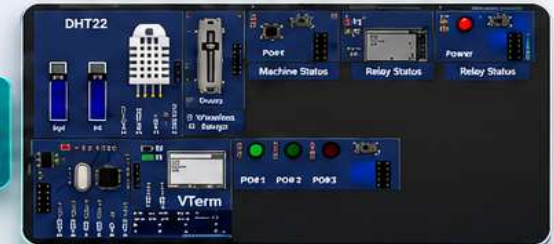
Status Indication (Normal/Warning/Alert)



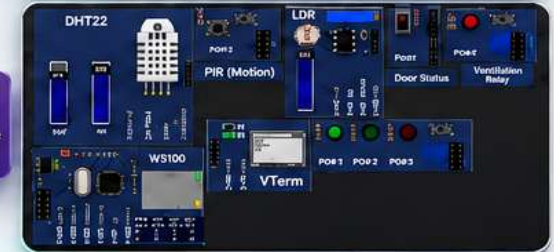
SIMULATION PLATFORM (PicsimLab)

Virtual simulation of two Arduino based nodes with sensors, actuators & network connectivity.

Node 1 Production Line



Node 2 Warehouse



SOFTWARE & TECHNOLOGIES



Arduino IDE

Code Development Environment



PicsimLab

Simulation & Virtual Hardware Platform



MQTT Protocol

Lightweight Pub/Sub Messaging Protocol



ThingsBoard Cloud

IoT Cloud Platform for Data Management



Dashboard & Analytics

Visualization, Trends & Alarm Management



Reliable

Industrial Grade Components



Real-Time

High Speed Data Acquisition



Scalable

Cloud Based Architecture



Smart Automation

Rule based Actuation



Secure

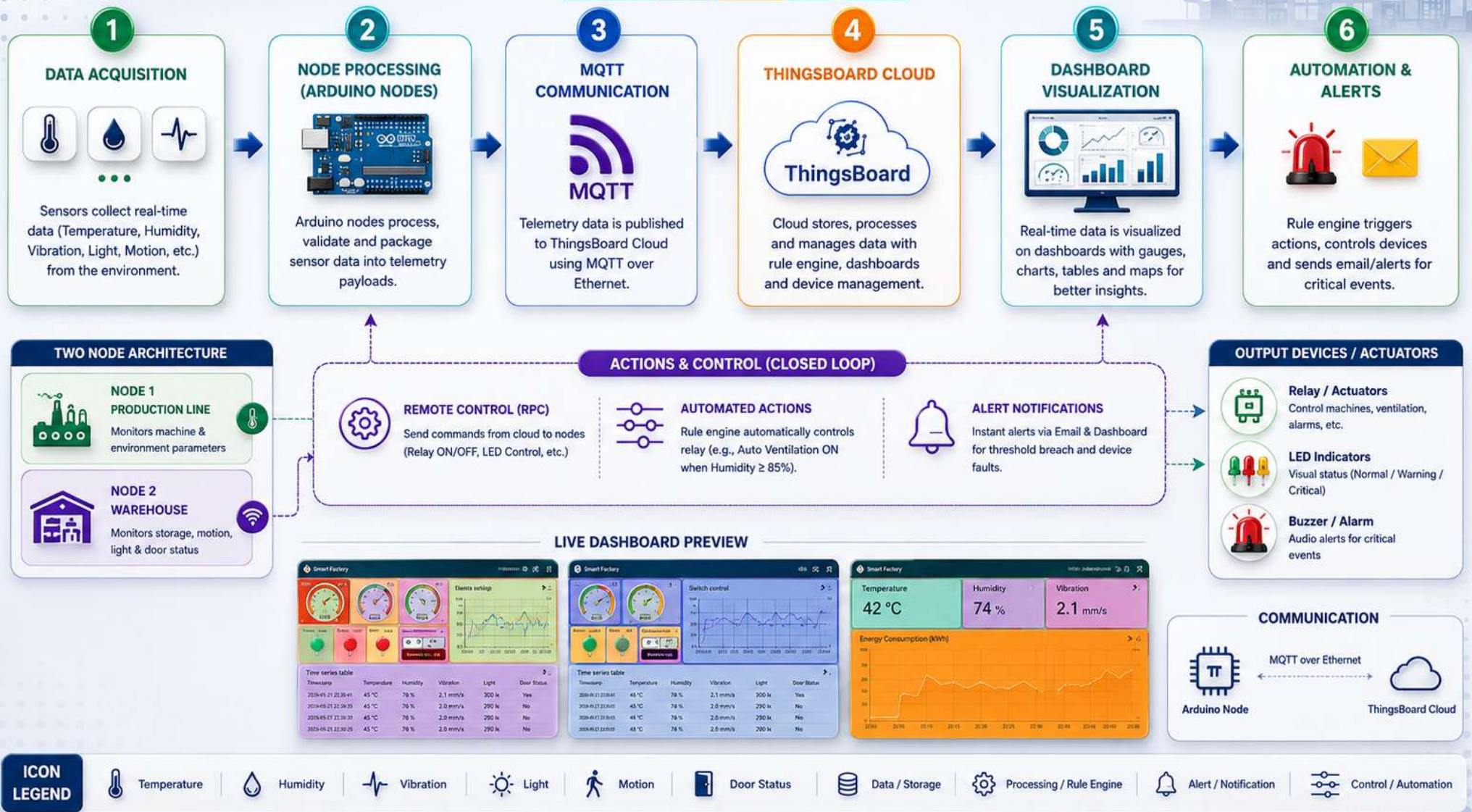
Encrypted Communication

TECHNOLOGY STACK

Cloud Platform	:	ThingsBoard Cloud
Communication	:	MQTT over Ethernet
Programming	:	Arduino (C/C++)
Data Storage	:	Cloud Database
Visualization	:	ThingsBoard Dashboards
Automation	:	Rule Engine & RPC

SYSTEM WORKFLOW

From Data Acquisition to Intelligent Automation



CONCLUSION & FUTURE SCOPE

Building a Smarter, Safer and More Efficient Industry with IoT



WHAT WE ACHIEVED



Real-Time Monitoring

Continuous monitoring of critical parameters across production line & warehouse.



Cloud Integration

Secure data transmission to ThingsBoard Cloud using MQTT over Ethernet.



Automation & Control

Rule based automation (e.g., Auto Ventilation) and remote control via RPC.



Instant Alerts

Real-time alarms and notifications for threshold breaches and events.



Dual Node Architecture

Monitoring two different environments (Production Line & Warehouse) simultaneously.



Data Visibility & Insights

Dashboards, charts and historical data for better decision making.



FUTURE SCOPE



AI / ML Integration

Predictive analytics for predictive maintenance and anomaly detection.



Mobile Application

Dedicated mobile app for real-time monitoring and remote control.



Edge Computing

Local data processing at edge for faster response and reduced latency.



Advanced Analytics

Deeper insights with advanced dashboards, KPIs and trend forecasting.



Enhanced Security

Stronger authentication, encryption and role-based access control.



Scalability

Expand to multi-site industrial ecosystem with more devices and sensors.



This project demonstrates how IoT and cloud technologies can transform traditional factories into **intelligent, connected and automated** environments.



Arduino Uno
(Nodes)



PicsimLab
(Simulation)



MQTT
(Protocol)



ThingsBoard Cloud
(Platform)



Dashboards
(Visualization)



Alerts & Notifications
(Real-Time)



Relays / Actuators
(Automation)



THANK YOU!



SENSE
Real-Time Data
from Sensors



CONNECT
Arduino Nodes &
MQTT Communication



ANALYZE
ThingsBoard Cloud
Processing



VISUALIZE
Dashboards &
Real-Time Insights



AUTOMATE
Smart Automation
& Alerts



PRESENTED BY
Rahul Sahu



A step towards **smarter** factories, **safer** operations and a **connected** future.